

DADAL, Mongolia

WMO: 442540

Lat: 49.0222N

Lon: 111.6117E

Elev: 2181

StdP: 77.72

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-28.2	-34.6	0.2	-28.5	-32.6	0.2	-27.1	7.4	-14.0	6.7	-13.8	1.4	270	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	30.2	16.0	27.9	15.6	25.9	15.2	18.6	25.5	17.6	23.9	16.7	23.0	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	15.6	21.2	15.7	14.6	20.4	14.7	13.7	19.5	62.9	25.7	59.2	23.5	56.0	23.2	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.0	7.0	6.0	-33.2	33.7	2.3	2.7	-34.9	35.6	-36.2	37.2	-37.5	38.8	-39.2	40.8		
			-33.5	20.3	2.4	1.3	-35.2	21.2	-36.6	22.0	-37.9	22.7	-39.6	23.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	0.7	-19.1	-15.5	-5.9	2.9	10.2	18.5	16.5	10.4	1.4	-10.0	-17.6	(6)	
(7)		DBStd	14.23	4.87	5.66	6.24	5.28	4.61	3.77	2.77	3.59	4.41	5.04	6.14	5.24	(7)
(8)		HDD10.0	4161	903	713	494	220	54	2	0	2	47	268	601	856	(8)
(9)		HDD18.3	6503	1161	946	752	463	256	89	30	77	239	524	851	1114	(9)
(10)		CDD10.0	779	0	0	0	7	59	183	264	204	59	2	0	0	(10)
(11)		CDD18.3	81	0	0	0	0	3	19	37	21	2	0	0	0	(11)
(12)		CDH23.3	1133	0	0	0	3	93	318	409	273	37	0	0	0	(12)
(13)		CDH26.7	335	0	0	0	0	25	103	126	74	7	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.2	2.4	2.6	3.1	3.3	2.5	2.1	2.2	2.4	2.4	2.2	2.2	(14)
(15)	Precipitation	PrecAvg	364	2	3	4	14	33	52	105	84	45	13	5	3	(15)
(16)		PrecMax	540	15	10	13	32	86	149	159	187	111	36	21	9	(16)
(17)		PrecMin	266	0	0	0	0	12	7	11	20	2	1	0	0	(17)
(18)		PrecStd	57	3	3	4	11	21	33	36	43	29	11	5	3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.9	2.3	14.2	22.6	29.6	33.1	34.1	31.8	27.6	18.9	7.9	-2.3	(19)
(20)			MCWB	-8.0	-2.9	3.8	9.0	12.3	16.2	18.5	17.7	13.8	8.7	1.8	-5.7	(20)
(21)		2%	DB	-7.8	-1.5	9.9	18.4	25.9	29.5	30.0	28.8	23.9	15.5	3.9	-5.8	(21)
(22)			MCWB	-10.0	-5.3	2.0	6.6	11.1	15.3	16.8	16.5	13.0	6.4	-0.5	-8.4	(22)
(23)		5%	DB	-10.1	-3.9	6.5	15.1	22.8	26.8	27.4	26.5	21.3	12.4	1.3	-8.2	(23)
(24)			MCWB	-11.9	-6.9	0.1	5.1	9.7	14.6	16.8	15.8	11.5	4.5	-2.5	-10.1	(24)
(25)		10%	DB	-12.0	-6.4	3.3	12.1	20.0	24.6	25.4	24.2	18.9	9.7	-0.8	-10.3	(25)
(26)			MCWB	-13.5	-9.0	-1.8	3.5	8.7	13.8	16.4	15.2	10.7	3.1	-3.9	-11.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-7.8	-2.8	4.1	9.5	13.1	18.5	20.4	20.0	15.9	8.9	2.2	-5.6	(27)
(28)			MCDWB	-5.3	1.8	12.8	21.8	26.8	29.1	29.1	28.0	24.5	17.1	7.8	-2.4	(28)
(29)		2%	WB	-10.0	-5.1	2.2	7.1	11.8	16.7	19.0	18.2	14.1	6.8	0.0	-8.1	(29)
(30)			MCDWB	-8.0	-1.5	9.5	17.6	24.0	25.8	25.5	24.8	19.6	14.8	4.2	-5.9	(30)
(31)		5%	WB	-11.8	-6.9	0.2	5.4	10.4	15.7	18.1	17.0	12.8	4.7	-1.6	-10.0	(31)
(32)			MCDWB	-10.2	-3.8	6.2	14.3	21.5	23.8	24.2	23.4	18.4	11.6	1.7	-8.2	(32)
(33)		10%	WB	-13.5	-8.8	-1.7	3.8	9.1	14.7	17.3	16.1	11.6	3.4	-3.2	-11.8	(33)
(34)			MCDWB	-12.1	-6.5	2.8	11.4	18.8	22.5	23.3	21.9	18.1	9.2	-0.2	-10.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	11.2	12.1	12.8	14.5	13.4	11.7	12.5	13.0	11.7	9.8	8.2	(35)
(36)			MCDBR	10.5	13.2	15.1	17.4	19.6	17.6	15.9	16.6	17.0	16.1	11.5	9.4	(36)
(37)			MCWBR	8.7	10.2	8.9	9.2	8.9	6.9	6.0	6.8	8.0	8.7	7.6	7.8	(37)
(38)		5% WB	MCDWB	10.2	13.1	14.5	16.5	17.7	14.6	12.5	12.9	13.8	14.9	11.1	9.4	(38)
(39)			MCWBR	8.6	10.1	8.8	9.0	8.7	6.2	5.6	6.0	6.9	8.5	7.7	7.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.203	0.220	0.248	0.295	0.335	0.340	0.320	0.300	0.269	0.241	0.217	0.204	(40)	
(41)		taud	2.439	2.436	2.455	2.402	2.306	2.382	2.483	2.525	2.575	2.583	2.510	2.429	(41)	
(42)		Ebn at Noon	918	970	984	957	923	918	934	942	949	934	891	866	(42)	
(43)		Edh at Noon	58	76	90	107	123	115	103	94	80	66	54	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.79	2.98	4.51	5.52	6.18	6.12	5.78	5.24	4.27	2.95	1.86	1.40	(44)	
(45)		RadStd	0.07	0.12	0.17	0.30	0.36	0.37	0.35	0.34	0.30	0.13	0.08	0.10	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (4 neighbors)	N/A	N/A	-1.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page